

Clinical Risk Assessment Report: Patient 11

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 10.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.48x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 3.2%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.18 [Avg (31th %ile)] (+0.5%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.87 [Bottom 18%] (-3.2%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Patient Age (Age) **Value: 65.0 [Avg (54th %ile)] (-1.2%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.48 [Avg (58th %ile)] (-1.2%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.45 [Avg (36th %ile)] (-0.7%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 1.0 [Top 18%] (-0.6%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

PET Radiomic Metabolic Feature (GLZLM_ZLNU.PET) **Value: -0.52 [Bottom <1%] (-0.5%)**
Metabolic PET feature reflecting glucose avidity and cellular metabolic rate.

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 0.2 [Avg (54th %ile)] (-0.5%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Machine Learning Feature Attribution (SHAP Decision Path)

